

## Education

### Bilkent University

Ankara, Turkey

Electrical-Electronics Engineering, Comprehensive Merit Scholarship. GPA: 3.58 January 2022-January 2026  
Relevant Coursework: Digital Electronics, Microprocessors, Analog Electronics, Electronic Circuit Design, Signals and Systems, Digital Signal Processing, Linear Algebra, Probability, Neural Networks, Machine Learning

### English Preparatory School

Ankara, Turkey

English Preparatory Course at Bilkent University.

September 2021 – December 2021

### Huseyin Avni Sozen Anatolian High School

Istanbul, Turkey

Graduated with 99/100.

July 2021

## Experience

### AlkanLab

Ankara, Turkey

#### Undergraduate Researcher

October 2024 – Present

- Bonito model implementation on Dorado base calling using PyTorch. Benchmarking on different processors, including PIM AI chips. Completed in scope of EU Project, BioPIM.
- Working on project "POACHA: Partial Order Alignment Computation Hardware Acceleration." FPGA and ASIC design for sequence-to-graph alignment with bit-vector representation.
- Supervised by Prof. Can Alkan

### ETH Zurich Integrated Systems Laboratory

Zurich, Switzerland

#### Undergraduate Summer Intern

June 2024 – September 2024

- Pre-training and fine-tuning 10 CNN and ViT models on multimodal skin cancer dataset.
- Pruning and quantization implementation for hardware-efficient AI. Deployed on RISC-V based GAP9 board.
- Analyzed deployment performances and accuracy results.

### Turkish Aerospace Industries

Ankara, Turkey

#### Summer Intern

August 2025 – September 2025

- Benchmarked YOLOv5 and YOLOv8 models that are being used in small-moving object detection in UAVs on Nvidia Jetson Orin GPU and DLA (Deep Learning Accelerator) cores.
- Completed block diagram of DPU (Deep Learning Processing Unit) for ZCU104 and ZCU106 boards using Vivado flow and configured PetaLinux project to extract SD Card Image. Deployed YOLO models on Xilinx UltraScale+ MPSoC ZCU106.

## Projects

### POACHA: Partial Order Alignment Computation Hardware Acceleration

June 2025 - Present

- FPGA and ASIC design implementation for partial order alignment (POA) based sequence-to-graph alignment.
- Dynamic programming by slice and bit-vector implementation on hardware.

### DELTIS: Deep Learning Thermal Image Super-Resolution

September 2024 - May 2025

- Collaboration with TUBITAK (Scientific and Technological Research Council of Turkey)
- To increase resolution of cheaper and compact IR cameras used in TUBITAK, a novel thermal image dataset is curated and a novel deep learning model is developed.
- Novel model and other three successful models existing in literature are embedded into an application that implements super-resolution for inputted image and prints various evaluation metrics.

## TRC-11

Spring 2023

- Completed the schematic, PCB design and implementation of transceiver (transmitter-receiver).
- It operates at 27 MHz, and has 11 m wavelength.

## Smart Home System

Fall 2023

- BASYS3 FPGA board based smart home system for temperature and motion detection.
- PIR motion sensor turns the LED on when motion is detected.
- LM35 analog sensor implements ADC and displays real time temperature with 7-segment display.
- Done on VHDL.

## Talks & Activities

### CGSI UCLA 2025

Los Angeles, CA

#### Participant and Presenter

July 2025 – August 2025

- Computational Genomics Summer Institute held in UCLA.
- Got accepted and participated to long program on-site, joined many networking activities.
- Presented my ongoing project POACHA.

### EFCL Summer School 2024

Zurich, Switzerland

#### Participant

June 2024

- ETH Zurich Future Computing Laboratory (EFCL) Summer School on Open Source IC Design and Computer Architectures.
- Participated in Track 3: Embedded AI for Bio signal Processing, being one of the 24 participants.
- Hands-on tutorial sessions with GAP9 for biosignal processing besides 18 hours theoretical lectures and networking sessions.

### XPONENTIAL 2024 - Global Uncrewed Systems and Robotics Conference

San Diego, CA

#### Attendee

April 2024

- Engaged in professionals from the leading companies and research groups in embedded and autonomous systems and robotics.
- Technical sessions and industry panels on developments in aerospace, defense, and commercial industries.
- Gained insights into interdisciplinary collaboration between academia, startups, and global companies.

## Awards & Achievements

### National University Entrance Exam

June 2021

- Ranked 71st among over 2.5M exam takers

### Merit-Based Scholarship by Council of Higher Education (CoHE)

- Awarded with merit-based scholarship by CoHE due to being one of the top 100 students. This scholarship covers monthly pocket money and continues during undergraduate education.

### Comprehensive Scholarship by Bilkent University

- Awarded with merit-based comprehensive scholarship by the university that covers all tuition, free dormitory accommodation and monthly pocket money during undergraduate education.

## Skills & Interests

**Technical:** Embedded C, C++, Java, Python, VHDL, Verilog, Vivado

**Language:** English: Professional and proficient. Turkish: Native. German: Beginner.

**Interests:** Formula 1, cars, tennis, sudoku, hiking, pilates, fitness